ABIRAMI R

Artificial Intelligence and Data Science Student
Coimbatore, Tamil Nadu | GitHub | LinkedIn | LeetCode

SUMMARY

Artificial Intelligence and Data Science student with hands-on experience in Python, Java, Machine Learning algorithms, Data Analysis, Data Structures and Algorithms, and Database Management Systems. Experienced in developing AI-based projects in astronomy and music classification. Passionate about Machine Learning and building practical, intelligent solutions to real-world problems.

EDUCATION

B.Tech – Artificial Intelligence and Data Science **2024–2028**
Bannari Amman Institute of Technology, Tamil Nadu **CGPA: 7.97/10**

TECHNICAL SKILLS

Programming: Python, Java
Machine Learning & AI: Machine Learning Algorithms, Feature Extraction, Audio Signal Processing
Libraries & Data Science: NumPy, Pandas, Seaborn, Skyfield, Librosa
Web & Development Tools: HTML, CSS, JavaScript, Git, GitHub, Figma
Core Computer Science: Data Structures and Algorithms, DBMS

PROJECTS

AI-Powered Stargazing Assistant

- Developed an intelligent astronomy assistant to help users explore visible celestial objects, constellations, planets, and astronomical events.
- Integrated astronomical data and interactive features to provide personalized stargazing guidance for beginners and astronomy enthusiasts.
- Used Skyfield and astronomy-based data processing for celestial object identification and visualization.

Audio Source Separation Using Demucs

- Developed a Python-based audio source separation application using the Demucs deep learning model to separate vocals and instrumental components from mixed audio recordings.
- Built an interactive Streamlit interface for audio uploads, source separation, playback, and downloading separated audio stems.
- Integrated deep learning-based audio processing to simplify music source separation through a user-friendly application.

Highlight Visible Constellations

- Developed a web-based stargazing application to help users identify and explore constellations visible in the night sky.
- Designed an interactive interface to present constellation information and support astronomy exploration.
- Focused on making celestial object discovery accessible through a simple and engaging web application.

CERTIFICATIONS

- Basics of Python – Infosys Springboard

ACHIEVEMENTS & HACKATHONS

- Won an inter-college competition at **Semmozi**, conducted by Kumaraguru Institute of Technology.
- Participated in **Bits and Build**, an online hackathon conducted by SRM Institute of Science and Technology.
- Participated in multiple technical and general hackathons, gaining experience in rapid prototyping, teamwork, and problem-solving.

SOFT SKILLS

Problem Solving, Communication, Teamwork, Adaptability, Time Management, Creativity, Leadership, Quick Learning

ADDITIONAL INTERESTS

Continuous Learning and Exploring New Technologies | Space and Astronomy | Carnatic Music and Singing | French Language – A1 Preparation